Problem 1:

$V_{GS} = 1.5V \xrightarrow{Sat} I_{DS} = 10\,\mu A$

$V_{GS} = 2.1V \xrightarrow{Sat} I_{DS} = 23.6\,\mu A$

a) $\frac{I_{DS_1}}{I_{DS_2}} = \frac{(V_{GS_1} - V_{th})^2}{(V_{GS_2} - V_{th})^2}$ $\therefore$ $\boxed{V_{th} = 0.5V}$

b) $I_1 = \frac{k}{2}(V_{GS_1} - V_{th})^2$ $\therefore$ $\boxed{k = 20\,\mu A/V^2}$

c) $k = \mu_n C_{ox} \frac{W}{L}$ $\therefore$ $\boxed{W = 260\text{ nm}}$

d) No, because we don't know V_{DS} which appears in the current equation in ohmic.

Problem 2:

$\boxed{2}$ From lecture:

$V_{OH} = V_{DD} - V_{th_L}$

$\therefore$ $\boxed{V_{OH} = 4.3V}$

$I_{D\,ohmic} = I_L|_{Sat}$

$k\left(\frac{W}{L}\right)_D \left[(V_{DD} - 2V_{th})V_{OL} - \frac{V_{OL}^2}{2}\right] = \frac{k\left(\frac{W}{L}\right)_L}{2}(V_{DD} - V_{th_L} - V_{th_D})^2$

$-0.5V_{OL}^2 + 12.2V_{OL} - 18.49 = 0$

$V_{OL} = 6.12V$ or $\boxed{V_{OL} = 2.01V}$

verify that M_D in ohmic:

$V_{OL} \overset{?}{<} V_{DD} - V_{th_L} - V_{th_D}$

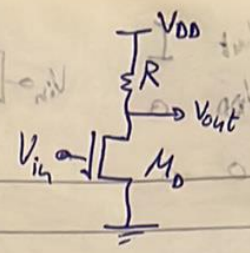
$2.01 < 3.6$ ✓

c) Increasing $\left(\frac{W}{L}\right)_D$ will not affect V_{OH} because the driver is cut-off. However, in case of V_{OL}, increasing $\left(\frac{W}{L}\right)_D$ will decrease the driver's resistance (remember: $R_{ON_{MOSFET}} = \frac{1}{K(V_{GS} - V_{TH})}$) and hence $\boldsymbol{V_{OL}}$ **will decrease.**

Problem 3.

$\boxed{3}$ $V_{th} = 0.4V$, $V_{DD} = 5V$

$L_{min} = 65\,nm$, $k' = 3.37\,\mu$



a) $\frac{W}{L} = 10$, $V_{OL} = 0.9V$, $R = ?$

logic '1' input:

$V_{in} = V_{DD}$ ∴ M_D on. Sat or ohmic? we want it ohmic. KCL:

$$I_D = I_R$$

$$k'\frac{W}{L}\left[(V_{DD}-V_{th})V_{OL} - \frac{V_{OL}^2}{2}\right] = \frac{V_{DD}-V_{OL}}{R}$$

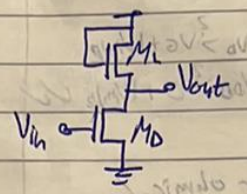
$$\cancel{V_{OL}^2\left[-\frac{1}{2}k'\frac{W}{L}\right] + V_{OL}\left[k'\frac{W}{L}(V_{DD}-V_{th}) + \frac{1}{R}\right] - \frac{V_{DD}}{R} = 0}$$

$$\cancel{4.685\times10^{-5}V_{OL}^2}\quad R = \frac{V_{DD}-V_{OL}}{k'\frac{W}{L}\left[(V_{DD}-V_{th})V_{OL} - \frac{V_{OL}^2}{2}\right]} \quad \therefore \boxed{R = 32.573\,k\Omega}$$

b) $Area = 1(\mu m)^2 \times k\Omega hm \times 32.573\,k\Omega hm = \boxed{32.573(\mu m)^2}$

c) $Area|_{total} = Area|_{M_D} + Area|_R = WL_{min} + Area|_R = 10L_{min}^2 + Area|_R \therefore \boxed{A|_{total} = 32.615(\mu m)^2}$

d) $V_{OH} = V_{DD} - V_{th}$



logic '1' input:

M_D on and assume in ohmic

$$I_L|_{sat} = I_D|_{ohmic}$$

$$\frac{\cancel{k'}\frac{W}{L})_L}{2}(\underset{5}{V_{DD}} - \underset{0.9}{V_{OL}} - \underset{0.4}{V_{th}})^2 = \cancel{k'}\frac{W}{L})_D\left[(\underset{5}{V_{DD}} - \underset{0.4}{V_{th}})(\underset{0.9}{V_{OL}}) - \frac{V_{OL}^2}{2}\right]$$

$$\frac{W}{L})_L = 4.93 \quad \therefore W_L = 4.93L_{min} \quad \therefore \boxed{W_L = 5L_{min} = 325\,nm}$$ (since can only be integer multiple)

$$A|_{total} = \sum WL = (W_L + W_D)L_{min} \quad \cancel{325nm + 10L_{min}} \quad \cancel{A|_{total}} = (5L_{min} + 10L_{min})L_{min} = 15L_{min}^2$$

$$\therefore \boxed{A|_{total} = 0.0633(\mu m)^2}$$

$< 32.615(\mu m)^2$

$A|_{total}$

Since want V_{OL} ↓ & W_D ↑ (to decrease driver's resistance) & W_L (L ... resistance)

∴ assume ... ∴ $3W_L \leq 501.7$... $\Rightarrow W_L = 167.256\,\mu m$

$W_D = 334.513\,\mu m$

but W_D & W_L should in... multiple...

$\frac{W_D}{L}$ ... $.35 \approx 5146 \Rightarrow W_D$ ... $334.49\,\mu m$

$\frac{W_L}{L_{min}} = 2573.1 \approx 2573 \Rightarrow W_L = 167.2$... μm

e) We are sacrifying area for the sake of decreasing V_{OL}.

For $\cancel{V_{OL}} \Rightarrow \cancel{\left(\frac{W}{L}\right)_D}$ & $\cancel{\left(\frac{W}{L}\right)_L}$

$A_{total} < 32.616(\mu m)^2$, $(WL)_D + (WL)_L < 32.616(\mu m)^2$

$k_D L_{min}$ L_{min} $k_L L_{min}$ L_{min}

$\left(\frac{W}{L}\right)_D$ $\left(\frac{W}{L}\right)_L$

$\therefore k_D + k_L < \frac{32.616(\mu m)^2}{L_{min}^2}$

$\therefore k_D + k_L < 7719.763$. Since k_D & k_L should be integers $\Rightarrow$ $k_D + k_L \leq 7719$

for minimum V_{OL}, choose $k_D = 7718$, $k_L = 1$

Calculate V_{OL} in terms of k_D & k_L :

$\left(\frac{W}{L}\right)_D$ $\left(\frac{W}{L}\right)_L$

$$1 \cancel{\frac{k_L}{2}}(\underbrace{V_{DD}}_{5} - V_{OL} - \underbrace{V_{tn}}_{0.4})^2 = \underbrace{k_D}_{7718}\left[(\underbrace{V_{DD}}_{5} - 2\underbrace{V_{tn}}_{0.4})V_{OL} - \frac{V_{OL}^2}{2}\right]$$

$$\frac{1}{2}(4.6 - V_{OL})^2 = 7718\left(4.2V_{OL} - \frac{V_{OL}^2}{2}\right)$$

$$3859.5\,V_{OL}^2 - 32{,}420.2\,V_{OL} + 10.58 = 0 \quad \text{so} \quad \boxed{V_{OL} = 0.3264\ mV}$$